



how to install
ConT_EXt

Introduction

Nowadays most T_EX users will use one of the many distributions that are available for free or commercially. This means that much of the hard work of installation is already taken care of. When installed properly, the user will quite certainly have the binaries, hyphenation patterns, fonts and some macro packages on the system.

This manual is therefore not targeted at installing T_EX, but focusses on how to get CON_TE_XT running within an existing distribution. When, after reading this text, you still cannot get CON_TE_XT working properly, we advice you to contact the maintainer of your distribution, or to send your questions to the CON_TE_XT mailing list.

Unpacking the archives

The CON_TE_XT distribution consists of several zipped archives. You need to unpack these to the appropriate directory of you file system. The archives are zipped using the free `zip` program, and can be unzipped using its counterpart.¹ Just type:

```
unzip -a <archive>.zip
```

Everything you need to get started can be found in the main CON_TE_XT archive:

`cont-tmf.zip` the CON_TE_XT sources and programs

There are some more archives you may want to install, like:

`cont-img.zip` a couple of extra resources

`cont-exa.zip` the EX_AM_PL_E framework tools and styles

There are two more somewhat redundant archives. These can be useful for users of packages other than CON_TE_XT. If you are using the files from `cont-tmf.zip`, it is not necessary to include the files from these archives.

`cont-ppc.zip` the PPCH_TE_X only files

`cont-mpd.zip` the METAPOST to PDF conversion tools

To prevent duplication in files, we strongly advise to obey the path as coded in the archives. The files in these archives will be unpacked into the appropriate directories of the official `texmf` tree. Some files are stored in the directories for generic T_EX files or the L^AT_EX package, but those are not important for CON_TE_XT users.

¹ Shareware programs like WinZip for MSWindows also work well.

<code>texmf/tex/context/config</code>	some configuration files
<code>texmf/tex/context/base</code>	all CON _T E _X T core files, modules
<code>texmf/tex/context/extra</code>	some goodies
<code>texmf/tex/context/sample</code>	a couple of sample files
<code>texmf/tex/context/user</code>	user specific files
<code>texmf/tex/context/third</code>	third party contributions
<code>texmf/metapost/context</code>	the METAPOST modules
<code>texmf/context/config</code>	some configuration files
<code>texmf/context/data</code>	data files used by scripts
<code>texmf/scripts/context</code>	the PERL and RUBY scripts
<code>texmf/doc/context/base</code>	CON _T E _X T documentation

Users can best not put files in the `base`, `extra` and `sample` directories. That way they can conveniently be removed and reinstalled. The files in `third` are to be organized by author.²

The way PERL and RUBY scripts are started differs per operating system. For instance, on UNIX, the PERL scripts should be installed without the `.pl` extension, because these scripts and possibly other programs rely on these names. They should be moved to the search path for binaries and scripts. On MS WINDOWS running scripts is not supported by the operating system, although in MS WINDOWS 2000 and later you can comfortably associate extensions (suffixes) with programs. Most script environment installers will set this association for you. In the binary tree of the T_EX distributions you can find stubs like `irun`, written by Fabrice Popineau. When copied to `<scriptname>.exe`, this program launches the script with the same name.

The initialization files for T_EXEXEC go into `texmf/context` (T_ET_EX) or into the same directory as the binaries. When setting up T_ET_EX, make sure you enable generation of the format files, by uncommenting the lines that specify the CON_TE_XT formats. You can edit the T_ET_EX (and F_PT_EX) configuration file using:

```
fmtutil
```

In T_ET_EX you should pass the switch `-edit`. When you pass `-all` you get all formats. When in need for patterns other than the default ones, `texconfig` can be used to enable more hyphenation patterns in the file `cont-usr.tex`. You can also edit this file directly.

Another way to deal with scripts is to use TEXMFSTART

² You need to create this (yet empty) directory yourself.

```
texmfstart --make --windows --indirect texexec.pl
texmfstart --make --unix      --indirect texexec.pl
```

The resulting files (`texexec` and/or `texexec.bat`) can be moved to your binary path. The advantage of this approach is that it is less sensitive to changes in the formal directory structure, since it will be maintained in a downward compatible way. You need to do the same for other tools, like `TEXUTIL`.

`CONTEXT` comes with a font that contains navigational symbols. These fonts go to the corresponding places in the fonts tree, in our case:

<code>texmf/fonts/tfm/hoekwater/context</code>	the files with suffix <code>tfm</code>
<code>texmf/fonts/afm/hoekwater/context</code>	the files with suffix <code>afm</code>
<code>texmf/fonts/type1/hoekwater/context</code>	the files with suffix <code>pfb</code>

There are also `CONTEXT` specific encoding/map files. These are stored in the following paths:

<code>texmf/fonts/map</code>	<code>CON_TE_XT</code> fontmap file
<code>texmf/fonts/enc</code>	<code>CON_TE_XT</code> font encoding file

In older distributions those files had to go under `texmf/dvips/config`. Unfortunately `TEX` distributions are not upward compatible in this respect.

Don't forget to add a reference to this file `context.map` to the `pdftex.cfg` file that resides in the `PDFTEX` configuration directory.

```
map +context.map
```

Instead you can configure `cont-sys.tex` (being your personal copy of `cont-sys.rme`) to let `CONTEXT` load map files on demand.

Setting up `TEXEXEC`

`TEXEXEC` is the command line interface to `CONTEXT`. There is nothing wrong with running `CONTEXT` in the traditional way, like

```
pdfetex &cont-en filename
```

or for a `WEB2C` based `TEX`

```
pdfetex &cont-en --programe=context filename
```

but, and this will be more clear when we provide more options, the next call is more convenient:

```
texexec filename
```

This not only hides the implementation specific switches, but also removes the need for specifying the format.

T_EXEXEC is written in PERL, a scripting language that is available on most leading software platforms. In order to operate well, we need to set up T_EXEXEC. Of course you must have PERL running on your system. On UNIX systems, this is always the case, but for MS WINDOWS you have to install PERL yourself.³

First you have to move T_EXEXEC and its relative T_EXUTIL to a location in the binaries path. When issuing the command `texexec` you should get some response. Even better, when saying:

```
texexec --verbose
```

you should get some information on how T_EXEXEC is set up. When generating formats and processing files fail, you need to set up the initialization file `texexec.ini`. This file comes disguised as `texexec.rme`, so when not present, you need to copy this file. The initialization file should be present in the `config` path, or in the same path as the script. Next you need to edit this file. The simplest way is to comment and uncomment one of the following lines:

```
set TeXShell to tetex
%set TeXShell to fptex
%set TeXShell to miktex
%set TeXShell to private
```

When this is done, you should check to what extent the rest of the variables in this file match the local settings. We hope that the names of the variables used are clear.

When it is not set up properly, T_EXEXEC tries very hard to locate the files it needs. Normally T_EXEXEC should start up rather quickly. When you are under the impression that you are waiting too long, there is probably an error in the setup.

Using T_EX

When you are using t_EX or derived distributions, you can usually stick to the regular updates, unless you want to use the latest version of CON_T_EX_T. In any event, you

³ A good choice is the Active State distribution that you can fetch from www.activestate.com

should make sure that only one copy is present on your system, because otherwise files can get mixed, due to the often aggressive file searching algorithms. If you want to update anyway, you can unzip `cont-tmf.zip` from within the `texmf` directory and regenerate the format files.

The PPCT_{EX} only archive is for users who maintain their own files and only want to install this package. The T_{EX}UTIL archive is for those who want to use this script, but don't want to install CON_TEX_T. Neither archive is needed when you install CON_TEX_T from `cont-tmf.zip`.

Using FPT_{EX}

The first step in installing CON_TEX_T under FPT_{EX}, is to unzip the file `cont-tmf.zip` within the `texmf` directory. Afterwards the `texexec` binaries and PERL script should be copied to the directories that contain the other T_{EX} binaries. The T_{EX} binaries path, that should also be part of your `PATH` variable, often looks like:

```
../tex/bin/win32
```

When no file `texexec.exe` is found, you have to unzip `texexec.zip` and copy `runperl.exe` to `texexec.exe`. Don't forget to update the `ls-R` file database by running `mktextlsr`.

Next you need to locate the file `texmf/web2c/fmtutil.cnf`. In this file, which contains documentation, you need to activate the CON_TEX_T formats. Now you can run:

```
texexec --make
```

Using MIK_TEX

This section is provided by Grzegorz Sapijaszko and Ed L. Cashin and concerns the installation of CON_TEX_T under MIK_TEX. Installation in the MIK_TEX environment isn't much different from the T_{EX} installation. Nevertheless, you should take a few steps to achieve good results. The first one is copying a `texexec.rme` file to `texexec.ini` and uncommenting the lines for MIK_TEX as follows:

```
%set TeXShell to tetex
%set TeXShell to fptex
set TeXShell to miktex
%set TeXShell to private
```

In next step you should add a string `\context\perltk` to your environment variable `PATH` (in `autoexec.bat` under WIN9X, or in Control Panel in WINNT), for example:

```
PATH=c:\miktex\context\perltk;
```

If you are using languages other than Dutch, German or English, you should uncomment the lines in the `cont-usr.tex` file for the hyphenation patterns of the languages you need. After that you can refresh the filename databases, for example, by using “Refresh Filename Databases” from the “Start|Programs|MikTeX|Maintenance” menu or “MikTeX|Options” menu, depending on the version you run. You should also generate a format file. This is described in the next section. Notice that you should have PERL installed on your system.

After generating the formats you should copy the format file `cont-xx.fmt` from `\context\perltk` to the directory where MikTeX is storing formats (usually `\localtexmf\miktex\fmt`). Another way is to add a `\context\perltk` directory to `miktex.ini` file:

```
[MikTeX]
...
...
;; Where MikTeX searches for .fmt files.
FMTPath=.;%R\miktex\fmt;C:\localtexmf\context\perltk//
```

As a convenience, you can copy the `runperl.exe` file from the `cont-wrk.zip` archive to `texexec.exe`. You should make sure sure that those binaries are in the same directory as the PERL scripts.

Generating formats

From its name you can deduce that `CONTEXT` is written in the typographic language `TEX`. `CONTEXT` is parameter driven, which means that users change its behaviour by setting variables and changing keys. `CONTEXT` comes with a multi-lingual interface. Currently there are three such interfaces: Dutch, English and German.

Users who want complete control can edit the file `cont-usr.tex` and generate a format using the main file `context.tex`. Users who want an Dutch, English or German format, can stick to the files named `cont-nl`, `cont-en`, and `cont-de`. Again, by editing the file `cont-usr.tex`, you can influence the outcome.

In the early years of `TEX`, generating a format was common practice and users were pretty well aware of format files, hyphenation patterns and fonts. Nowadays, distributions take care of the more complicated issues, so users can comfortably skip many

nasty installation steps. To make life even more comfortable, `CONTEXT` comes with `TEXEXEC`, a command line interface to `TEX`. When properly set up, this PERL script can save you much time.

For instance, generating the three formats mentioned is accomplished by:

```
texexec --make en de nl
```

When `TEXEXEC` is set up properly, this command should work. Before you read on, you should try to generate at least the English and Dutch format. The Dutch format is needed by some of the postprocessing features built into `TEXEXEC`.

```
texexec --make en nl
```

Additionally, you need to make the `MetaFun` format. This is an extension to the standard `METAPOST` format which offers not only more features, but also interfaces quite well to `CONTEXT`.

```
texexec --make metafun
```

The formats associated with the interfaces default to the language of the interface. This is all right for Dutch, English or German users, but Polish and Czech users are worse off. For them a format file that defaults to their own language makes more sense. Poles will like to say:

```
texexec --make --language=pl --bodyfont=plr en
```

while Czech people will go for:

```
texexec --make --language=cz --bodyfont=csr en
```

Both produce a format called `cont-en` with an English interface, but the first one defaults to Polish hyphenation patterns and fonts, and the second one to Czech ones. If wanted, you may pass a comma separated list of languages.

```
texexec --make --language=pl,it,uk --bodyfont=plr en
```

or, to generate a english interface format with Czech and Slovak patterns and Czech-Slovak Computer Modern Roman fonts:

```
texexec --make --language=cz,sk,en --bodyfont=csr en
```

At `PRAGMA ADE` we simply load all the languages:

```
texexec --alone --make --all
```

Unfortunately, the hyphenation patterns are of hard coded in a format file and cannot be loaded at run time. When patterns are needed other than the ones loaded by default, you can consider adapting the file `cont-usr.tex`. This file is loaded at format generation time. When for instance Italian patterns are to be used, given that these are available either in the file `lang-it.pat`, or in a file onto which this filename is mapped, you should uncomment the line:

```
\installlanguage [\s!it] [\c!state=\v!start] % italian
```

The strange looking `\s!` and `\c!` things are needed in order to support multiple interfaces. Don't touch these!

When using WEB2C, in `texmf.cnf` some CON_TE_XT specific memory settings take place. When directly generating a format —i.e. when you're not using the T_E_X initialization script or T_E_XEXEC— you should supply the program name: `-progrname=context`

Make sure you read the manual to T_E_XEXEC. Apart from the normal processing of files, there are quite a few useful options: mode dependant processing, output selection, generating booklets, typesetting contact sheets of figures, manipulating PDF files, and more.

Changing defaults

The somewhat more run-time specific settings, like certain special drivers, can be added to `cont-sys.tex`. This file is loaded at run time. For instance, this file can contain the line:

```
\setupoutput[pdftex]
```

This commands tells CON_TE_XT to produce PDF output by default. For Y&Y and Acrobat support, you just say:

```
\setupoutput[dviwindo,acrobat]
```

Of course you can also load location specific layout settings in this file. The next few lines tell CON_TE_XT to default to the CON_TE_XT navigational symbols, instead of the ones composed from other glyphs.

```
\usesymbols [nav]
\setupsymbolset [navigation 1]
```

At PRAGMA ADE we want to process METAPOST files at run-time, so there we also have entries like:

```
\runMPgraphicstrue
\runMPTEXgraphicstrue
\useMETAFUNformattrue
```

The first setting leads to less intermediate files. The next two settings let CONTEX_T process the graphics real-time. For that you need to enable `write18` in the T_EX configuration file (for WEB2C this is `texmf.cnf`). The last setting only makes sense if you have also generated a *MetaFun* format. If you want to know more about *MetaFun*, take a look at its manual.

The verbatim environments provide pretty printing. When you want even more fancy verbatim, for instance with in-between switching of a language interpreter, you should enable this feature. If you get no idea what we're talking about here, leave this variable untouched since tricky code is involved.

```
\newprettytrue
```

When CONTEX_T cannot determine the dimensions of an external figure, and no `texutil.tuf` file is present, you can let CONTEX_T call T_EXUTIL directly. If you use PDF_T_EX, you can leave this switch off.

```
\runutilityfiletrue
```

More than one instance of T_EX a single path, can lead to clashes in temporary files. The next switch enables a filename security feature:

```
\protectbuffertrue
```

For the moment, we use these low level boolean switches instead of more readable commands.

An important section in this file are the font defaults. You can set a default font encoding as well as enable map file loading for PDF_T_EX. Some of these features are still experimental.

The WEB2C configuration

Although not strictly needed, CONTEX_T will operate more smoothly when these features are set in the file `texmf.cnf`.

```
openout_any = a
shell_escape = t
allow_multiple_suffixes = f
```

The first line permits $\text{CON}\text{T}_{\text{E}}\text{X}$ T to open parent paths that can hold common styles. The second line enables users to embed METAPOST code in their document. As a result, METAPOST is called automatically.⁴ The last line makes sure that when opening files like `somefile.tuo`, $\text{T}_{\text{E}}\text{X}$ will not try to open `somefile.tuo.tex` first.

When you embed $\text{T}_{\text{E}}\text{X}$ code in a METAPOST definition, using `btex ... etex`, the next line will use $\text{T}_{\text{E}}\text{XEXEC}$ to process that fragment in the case of using METAPOST directly.

```
MPXCOMMAND=texexec --mptex
```

However, the latest distributions don't need that kind of tweaking any more since $\text{T}_{\text{E}}\text{XEXEC}$ sort things out itself.

Processing files

You can test you installation with the following file.

```
\starttext
  \framed {Let's see if it works.}
\stoptext
```

If $\text{CON}\text{T}_{\text{E}}\text{X}$ T is set up properly, then the command

```
texexec filename
```

will produce a file `filename.dvi`. Unless a file was already processed before, you will notice that $\text{T}_{\text{E}}\text{XEXEC}$ processes the file at least two times. During a $\text{T}_{\text{E}}\text{X}$ run, $\text{CON}\text{T}_{\text{E}}\text{X}$ T saves information in the files `filename.tui`: cross references, entries to the table of contents, data needed for optimization, etc. If the run is succesful, this file is converted to a file called `filename.tuo` and used in the next run. $\text{T}_{\text{E}}\text{XEXEC}$ will reprocess the file until the `tuo` file is unchanged.

We *strongly recommend* to use $\text{PDF}\text{T}_{\text{E}}\text{X}$: Peter Breitenlohn's $\varepsilon\text{-T}_{\text{E}}\text{X}$ permits $\text{CON}\text{T}_{\text{E}}\text{X}$ T to run more efficient, while Hàn Thê Thành's $\text{PDF}\text{T}_{\text{E}}\text{X}$ provides PDF output and $\text{PDF}\text{T}_{\text{E}}\text{X}$ combined both in one program. For those unfamiliar with $\varepsilon\text{-T}_{\text{E}}\text{X}$: this is an extension of $\text{T}_{\text{E}}\text{X}$ that is not only more efficient and in some aspects faster, but that also offers

⁴ You can disable this feature by running $\text{T}_{\text{E}}\text{XEXEC}$ with the switch `-automp`.

a few features that come in handy when writing complicated code. For the moment, `CONTEX` adapts its behaviour to the kind of `TEX` that is used, but future versions may rely on `ε-TEX` or other descendants completely.

By default, `CONTEX` generates DVI output for DVIPS, unless the output is specified otherwise. We already mentioned the `\setupoutput` command. A second way of achieving this is:

```
texexec --pdf filename
```

And yet another way is adding a comment line in the document source, like:

```
% interface=en output=pdfTeX translate=cp1250p1
```

or

```
% interface=en output=pdfTeX translate=cp1250cz
```

Now we can omit the `-pdf` switch when we launch `TEXEXEC`. Normally `TEXEXEC` is able to sort out the interface itself, but in case of troubles, you can set some defaults in the file `texexec.ini`. The `translate` key is only needed when you use the reencoding-on-the-fly feature of `WEB2C`.

The distributions has a few demo files. You may try:

```
texexec --pdf demo-tex.tex
texexec --pdf demo-mps.tex
texexec --pdf demo-xml.tex
```

Minimal `CONTEX` trees

From the PRAGMA ADE website you can download a minimal `CONTEX` distribution. This is just a subset of `TEXLIVE`. On MS WINDOWS, this works as follows:

```
mkdir c:\context
cd /d c:\context
unzip mswintex.zip
setuptex
texexec --make --alone --all
```

You can now (whenever needed) initialize the tree with:

```
c:\context\setuptex.bat c:\context
```

Normally you will do this when a shell (cmd) is opened. There is no need to do additional configurations.

On UNIX you can do something:

```
mkdir /home/myself/context
cd /home/myself/context
unzip mslinux.zip
. setuptex
texexec --make --alone --all
```

Here you can initialize the tree with:

```
. /home/myself/context/setuptex /home/myself/context
```

When you don't provide a path, this script will try to figure out things itself. Since the tree is completely isolated from other installations, you can comfortably use multiple trees alongside.

Subscribing to the list

There are two dedicated mailing lists hosted by the NTG:

the CON_TE_XT mailing list
the PPCH_TE_X mailing list

These lists are so called *mailman* lists. You can subscribe yourself to a list using the web interface:

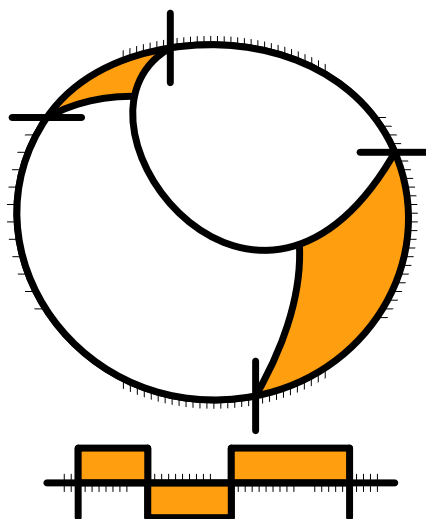
```
http://www.ntg.nl/mailman/listinfo/ntg-context
```

More information can be found at:

```
http://www.pragma-ade.com
```

Another source, rapidly growing in size and resources, is:

```
http://contextgarden.net
```



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